

The Accelerating Universe as Statistical Inference: Type Ia Supernovae, Maximum Likelihood, and the Cosmological Constant, 1988–1999

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Abstract

In their Nobel Prize–winning papers, Adam Riess, Brian Schmidt, and their High-Z Supernova Search Team [Riess et al., 1998], and independently Saul Perlmutter and the Supernova Cosmology Project [Perlmutter et al., 1999], reported that the expansion of the universe is *accelerating*. Their argument had two components. A physical *model* supplied the Friedmann equations with a cosmological constant Λ . A *measurement equation* related the apparent magnitude of a Type Ia supernova to its redshift, with the shape of that relation controlled by two density parameters Ω_M and Ω_Λ . This note examines their reasoning from a modern statistical perspective. I describe the data both groups assembled, the Calán/Tololo low-redshift training sample and the high-redshift discoveries at $z = 0.18$ – 0.83 . I then lay out the maximum likelihood problem they solved: choose $(\Omega_M, \Omega_\Lambda)$ to minimise the weighted sum of squared residuals between observed

*In 2011 my good friend Christopher A. Sims and I shared the Nobel Prize in economics. That good luck let us meet and become friends with winners of prizes in physics and other fields. Our conversations with the physics winners were especially interesting. Adam Riess had majored partly in economics as an undergraduate and told us about his very interesting undergraduate thesis. I have kept in contact with Adam Riess and Brian Schmidt, who have taught me the nuts and bolts of the computer programs that they used to compute and climb the likelihood function that led to their remarkable discovery. This paper is one of five that read episodes in the history of physics and astronomy as exercises in statistical inference and machine learning. Sargent [2025] sets out the theme: what is now called artificial intelligence names practices that scientists have used for centuries. The other four treat Kepler and Newton, Einstein’s photoelectric equation, Pauli’s exclusion principle, and the exoplanet discovery of Mayor and Queloz.

distance moduli and the Friedmann prediction. Under Gaussian errors MLE coincides with weighted least squares. The curvature of the log-likelihood surface—the Fisher information—determines the confidence ellipses in the $(\Omega_M, \Omega_\Lambda)$ plane. A nuisance parameter, the combination $\mathcal{M}$ of the absolute magnitude M_B and the Hubble constant H_0 , enters the measurement equation as the contact electromotive force enters Millikan’s photoelectric regression. It shifts the Hubble diagram vertically and leaves the shape of $\mu(z)$ alone, so the curvature of the diagram identifies the shape parameters independently of $\mathcal{M}$. The analogy stops there. Millikan’s slope was identified; these two parameters are identified only in one combination. The estimates under flatness, $\Omega_M \approx 0.28$ and $\Omega_\Lambda \approx 0.72$, imply a deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda \approx -0.58$.

I compute the Fisher information for the 1998 designs. It reproduces the published error bar on the combination $0.8\Omega_M - 0.6\Omega_\Lambda$, ± 0.1 , from the redshift distributions and quoted uncertainties alone. It also shows that the individual parameters are not measured: the marginal standard errors are 0.44 and 0.65. The published ± 0.09 is the standard error under the flatness constraint. Ruling out $\Omega_\Lambda = 0$ therefore requires a prior, either $\Omega_M \geq 0$ or flatness from the cosmic microwave background.

I read the episode in the language of machine learning. The Friedmann equations supplied the hypothesis class. The Phillips width–luminosity relation supplied the feature that made the signal visible, as Lenard’s finding that frequency rather than intensity governs photoemission supplied Einstein’s. A deep network given the same data recovers neither the density parameters nor their meaning. The supernovae measure one function, $d_L(z)$, and every theory reproducing it remains standing: theory reduces the sample size that data must supply, and theory also names the measurement that discriminates.

1 Introduction

In 1998 two independent teams reported that distant Type Ia supernovae are systematically *fainter* than they would be in a universe whose expansion is decelerating. [Riess et al. \[1998\]](#) appeared in September 1998 and [Perlmutter et al. \[1999\]](#) in June 1999; both groups had announced preliminary results at meetings earlier in 1998. The simplest explanation is that the expansion is speeding up, driven by a positive cosmological constant Λ or some analogous dark energy. Saul Perlmutter received one half of the 2011 Nobel Prize in Physics for the discovery, and Adam Riess and Brian Schmidt

shared the other half.

A twenty-first-century statistician confronted with tables of (redshift, apparent magnitude) pairs would recognise the problem as a weighted nonlinear regression with two scientifically meaningful parameters and a nuisance intercept. The Friedmann equations with a cosmological constant were well established before the data were collected. The discovery consisted in determining the sign and magnitude of Ω_Λ .

Six features of the episode interest a statistician.

1. **Model specification: the hypothesis class from the Friedmann equations.** General relativity predicts the luminosity-distance–redshift relation $d_L(z; \Omega_M, \Omega_\Lambda)$ exactly. The functional form is fixed, and the estimation problem reduces to two dimensionless density parameters.
2. **Standardising a noisy standard candle.** The peak B -band luminosities of raw Type Ia supernovae scatter by ≈ 0.4 magnitudes. [Phillips \[1993\]](#) found that the rate of post-peak decline Δm_{15} predicts peak luminosity, which reduces the dispersion to ≈ 0.15 mag.
3. **The nuisance parameter $\mathcal{M}$.** The combination of absolute magnitude M_B and Hubble constant H_0 shifts the entire Hubble diagram vertically. The observations do not identify $\mathcal{M}$ on its own, and both teams profiled it out of the likelihood. The relative distance moduli of high- z and low- z supernovae identify the shape of $\mu(z)$.
4. **Partial identification of the density parameters.** The shape of $\mu(z)$ pins down one combination of Ω_M and Ω_Λ and leaves the orthogonal combination nearly free. The published error bars assume flatness. Acceleration follows from the supernova data together with a prior, not from the supernova data alone.
5. **Independent replication with different methods.** The High-Z Supernova Search Team used the MLCS (Multi-Light-Curve Shape) standardisation method of [Riess et al. \[1996\]](#); the Supernova Cosmology Project used the stretch-factor method of [Perlmutter et al. \[1997\]](#). The two groups analysed different supernovae and reached the same conclusion to within their measurement uncertainties.
6. **Reluctance to accept the model.** Many physicists doubted $\Omega_\Lambda > 0$ because the prevailing theory held that the vacuum energy density should be negligible

[Weinberg, 1989].

Both teams had roughly 40–60 high-quality supernovae, a meagre dataset by modern standards. The Friedmann equations specified the regression function up to three free parameters and so collapsed a high-dimensional function-approximation task into a low-dimensional one. Even that proved too large for the data. Fifty supernovae determine one combination of the two density parameters and leave the other open, and the published estimates close it with a prior.

Sections 2–4 set the observational and theoretical stage. Section 5 presents the WLS/MLE procedure. Section 6 develops the Fisher information and confidence ellipses. Section 7 discusses the $\mathcal{M}$ identification problem. Section 8 presents the main results and the inference of acceleration. Section 9 compares the two teams. Section 10 describes the scepticism that greeted the result and the robustness tests that answered it. Section 11 describes the prediction that supernovae at $z > 1$ later confirmed. Section 12 describes what the supernovae could not settle. Sections 13–14 reinterpret the episode in the language of machine learning. Section 15 concludes.

A companion paper treats Millikan’s 1916 photoelectric experiment in the same way, and the comparison runs through what follows. Einstein’s equation is a line whose slope is a universal constant, fixed in advance by Planck’s blackbody fit and predicted with no free parameter. A contact electromotive force shifted every reading by the same amount and left the slope untouched, as $\mathcal{M}$ shifts every point of the Hubble diagram. Millikan measured the slope with five observations to better than one percent. The disanalogy matters more than the analogy: Millikan had one structural parameter, so removing the nuisance left it identified, while the Friedmann model has two and removing the nuisance leaves a second degeneracy behind. Section 6 turns on that difference.

2 Observational Background, 1929–1994

2.1 Hubble’s Law and the Standard Model of Cosmology

In 1929 Edwin Hubble [1929] reported that the recession velocities v of nearby “extra-galactic nebulae” satisfy

$$v = H_0 d, \tag{1}$$

where d is the distance and H_0 the Hubble constant. Hubble estimated $H_0 \approx 500 \text{ km s}^{-1} \text{ Mpc}^{-1}$; subsequent Cepheid-variable distance rescaling reduced the accepted value to $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Alexander Friedmann had already shown in 1922 and 1924 [Friedmann, 1922, 1924] that expanding cosmological solutions follow from Einstein’s field equations; Hubble’s measurement confirmed that we live in one of them.

In the Friedmann model, the dynamical variable is the scale factor $a(t)$, normalised so that $a(t_0) = 1$ today. The recession redshift z of a source satisfies $1 + z = 1/a(t_{\text{emit}})$. Three dimensionless density parameters characterise the matter-energy content:

$$\Omega_M = \frac{8\pi G \rho_{M,0}}{3H_0^2}, \quad (\text{non-relativistic matter}) \quad (2)$$

$$\Omega_\Lambda = \frac{\Lambda}{3H_0^2}, \quad (\text{cosmological constant / dark energy}) \quad (3)$$

$$\Omega_k = \frac{-k}{a_0^2 H_0^2}, \quad (\text{spatial curvature}) \quad (4)$$

with the constraint $\Omega_M + \Omega_\Lambda + \Omega_k = 1$. A flat universe has $\Omega_k = 0$, i.e., $\Omega_M + \Omega_\Lambda = 1$.

Einstein had introduced $\Lambda > 0$ in 1917 [1917] to obtain a static universe, then abandoned it after Hubble established expansion. The two teams’ discovery that $\Omega_\Lambda \approx 0.72$ rehabilitated the cosmological constant as an observational reality.

2.2 Type Ia Supernovae as Standard Candles

A *standard candle* is an astronomical object of known intrinsic luminosity. If absolute luminosity L is known, the observed flux $F = L/(4\pi d_L^2)$ gives the luminosity distance d_L directly. Comparing d_L with the redshift z maps the distance-redshift relation and thereby constrains the Friedmann parameters.

Type Ia supernovae are thermonuclear explosions of carbon-oxygen white dwarfs in binary systems [Filippenko, 1997]. The account current in the 1990s had the white dwarf accreting until it approached the Chandrasekhar limit $M_{\text{Ch}} \approx 1.4 M_\odot$, so that a near-universal exploding mass produced a near-universal luminosity. That account no longer commands consensus. Sub-Chandrasekhar double detonations and white-dwarf mergers are now taken seriously, and the progenitor population may well be mixed.

The uniformity of Type Ia peak luminosity is therefore an empirical fact rather than a

derived one, and it was empirical in 1998 too. Before standardisation the scatter in peak B -band absolute magnitude is $\sigma_{M_B} \approx 0.4$ mag. Since five magnitudes is a factor of one hundred in flux, that is a 20% distance uncertainty, too large for cosmology.

2.3 The Width-Luminosity Relation: The Key Standardisation Insight

Phillips [1993] established a tight empirical relation between the peak luminosity of a Type Ia supernova and the speed at which its light curve declines after maximum light. The parameter $\Delta m_{15}(B)$ measures the drop in B -band magnitude during the 15 days after peak: brighter (more energetic) supernovae have slower, broader light curves (Δm_{15} smaller), while fainter events decline more rapidly. Quantitatively, the relation is approximately

$$M_B^{\text{peak}} \approx M_B^{\text{ref}} + \alpha(\Delta m_{15} - 1.1), \quad \alpha \approx 0.75 \text{ mag}, \quad (5)$$

where M_B^{ref} is the absolute magnitude of a “normal” event with $\Delta m_{15} = 1.1$. The coefficient $\alpha \approx 0.75$ is the Hamuy et al. [1996] calibration. Phillips [1993] himself fitted a steeper slope, near 2.7 mag per unit $\Delta m_{15}(B)$, to nine supernovae; later samples flattened it. Both figures come from secondary sources and should be checked against the originals.

After the correction the intrinsic dispersion falls to $\sigma_{\text{int}} \approx 0.15$ mag. Published values run from 0.13 to 0.17 mag depending on sample and method, and the two teams quote 0.13 and 0.14. I use 0.15 mag throughout for the intrinsic dispersion and 0.22 mag for the total per-supernova error at $z \approx 0.5$, which adds photometric and K -correction terms.

Without the correction the scatter masks the cosmological signal. With it the $\mu(z)$ residuals separate a decelerating from an accelerating universe.

The Phillips relation is often called the supernova counterpart of Lenard’s discovery that the photoelectric stopping potential depends on frequency and not intensity. The two findings did different work. Frequency and intensity were both physical primitives, and Lenard’s experiment chose between candidates already on the table. Nothing put Δm_{15} on the table. Phillips found it by fitting curves to light curves, no theory predicted it beforehand, and none derives it now. Section 13.4 takes up what that difference means

for the reading of the episode offered here.

2.4 The Two Teams

The **Supernova Cosmology Project** (SCP), led by Saul Perlmutter at Lawrence Berkeley National Laboratory, began systematic high- z supernova searches in the late 1980s and developed a systematic discovery pipeline using ground-based telescopes (Isaac Newton Telescope, Cerro Tololo Inter-American Observatory, Keck). [Goobar and Perlmutter \[1995\]](#) describe their “batch mode” scheduling. The team discovered candidates in a first observing run and followed them up in a prescheduled second run three weeks later. The schedule guaranteed that the events were still near peak brightness when the spectroscopy was taken. By 1997 they had reported results for seven high- z supernovae [[Perlmutter et al., 1997](#)]. Their definitive 42-supernova paper appeared in 1999 [[Perlmutter et al., 1999](#)].

The **High-Z Supernova Search Team** (HZT), assembled around Brian Schmidt at the Mount Stromlo Observatory and Robert Kirshner at Harvard, made complementary use of the Hubble Space Telescope and larger ground-based facilities to obtain high-quality light curves and spectra. Adam Riess developed the MLCS analysis pipeline for the team’s data [[Riess et al., 1996](#)]. A first report on four HST supernovae [[Garnavich et al., 1998](#)] was followed by the definitive 16-plus-34 SN paper of [Riess et al. \[1998\]](#).

3 The Cosmological Model and the Measurement Equation

3.1 The Friedmann Equation

The first Friedmann equation with matter density ρ_M , pressure-free dust, and cosmological constant Λ is

$$H^2(z) \equiv \left(\frac{\dot{a}}{a}\right)^2 = H_0^2 \left[\Omega_M (1+z)^3 + \Omega_k (1+z)^2 + \Omega_\Lambda \right] \equiv H_0^2 E^2(z), \quad (6)$$

where z is evaluated on the past light cone and the present-day normalisation $E(0) = 1$ implies $\Omega_M + \Omega_k + \Omega_\Lambda = 1$. The dimensionless Hubble function $E(z)$ gives the expansion rate at redshift z in units of H_0 . For $\Omega_\Lambda = 0$ and matter-only content, $E(z) = (1+z)^{3/2}$

(the Einstein-de Sitter solution); for the best-fit flat Λ CDM model, $E(z)$ passes through intermediate values between these extremes.

3.2 Luminosity Distance

The luminosity distance to a source at redshift z is

$$d_L(z; \Omega_M, \Omega_\Lambda) = \frac{c(1+z)}{H_0 \sqrt{|\Omega_k|}} f\left(\sqrt{|\Omega_k|} \int_0^z \frac{dz'}{E(z')}\right), \quad (7)$$

where

$$f(x) = \begin{cases} \sin x & \Omega_k < 0 \text{ (closed)}, \\ x & \Omega_k = 0 \text{ (flat)}, \\ \sinh x & \Omega_k > 0 \text{ (open)}. \end{cases} \quad (8)$$

For the flat case of primary interest ($\Omega_k = 0$, $\Omega_M + \Omega_\Lambda = 1$), equation (7) reduces to

$$d_L(z; \Omega_M) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_M(1+z')^3 + (1-\Omega_M)}}. \quad (9)$$

Here $\Omega_\Lambda = 1 - \Omega_M$ is not a free parameter once flatness is imposed. For the general (non-flat) case, Ω_M and Ω_Λ are estimated jointly without the flatness constraint.

3.3 The Distance Modulus and Measurement Equation

The *distance modulus* is defined as

$$\mu \equiv m_B - M_B = 5 \log_{10} \left(\frac{d_L}{10 \text{ pc}} \right), \quad (10)$$

where m_B is the observed peak apparent magnitude (standardised for light-curve shape) and M_B is the intrinsic absolute magnitude. Since $1 \text{ Mpc} = 10^6 \text{ pc}$, this becomes

$$\mu = 5 \log_{10} \left(\frac{d_L}{\text{Mpc}} \right) + 25. \quad (11)$$

The theoretical distance modulus predicted by the cosmological model is therefore

$$\mu^{\text{th}}(z; \Omega_M, \Omega_\Lambda, H_0) = 5 \log_{10} \left(\frac{d_L(z; \Omega_M, \Omega_\Lambda)}{\text{Mpc}} \right) + 25. \quad (12)$$

The full measurement equation—the fundamental regression of the entire analysis—is

$$m_{B,i} = M_B + \mu^{\text{th}}(z_i; \Omega_M, \Omega_\Lambda, H_0) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2), \quad (13)$$

The sum of squared errors defines the likelihood. Four quantities appear on the right, M_B , H_0 , Ω_M and Ω_Λ , but only three combinations of them are estimable, as the next subsection shows.

3.4 The Nuisance Parameter $\mathcal{M}$

Expanding equation (12) reveals that H_0 and M_B enter only through a single combination. Write

$$d_L(z; \Omega_M, \Omega_\Lambda, H_0) = \frac{c}{H_0} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda), \quad (14)$$

where $\tilde{d}_L$ is the *dimensionless* luminosity distance that depends only on the density parameters. Then

$$\begin{aligned} \mu^{\text{th}} &= 5 \log_{10} \left(\frac{c}{H_0} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda) \right) + 25 \\ &= 5 \log_{10} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda) + 5 \log_{10} \left(\frac{c}{H_0 \text{ Mpc}} \right) + 25, \end{aligned} \quad (15)$$

so the measurement equation (13) becomes

$$m_{B,i} = \underbrace{M_B + 5 \log_{10} \left(\frac{c}{H_0 \text{ Mpc}} \right) + 25}_{\mathcal{M}} + 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda) + \varepsilon_i. \quad (16)$$

The scalar $\mathcal{M}$ shifts the Hubble diagram vertically; the term $5 \log_{10} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda)$ determines its shape and does not depend on $\mathcal{M}$.

- $\mathcal{M}$ is not identified by the shape of the Hubble diagram. Any vertical shift of the entire m_B sequence can be absorbed into $\mathcal{M}$.
- The *curvature* of $\mu(z)$ —the way the slope of the m_B - z diagram changes between low- z and high- z supernovae—is informative about Ω_M and Ω_Λ , and carries no information about $\mathcal{M}$.

Curvature separates the shape parameters from $\mathcal{M}$. It does not separate them from

each other. Over the observed redshift range a rise in Ω_M and a rise in Ω_Λ bend $\mu(z)$ in nearly opposite ways, so the data fix one combination of the two and leave the orthogonal combination nearly free. Section 6 quantifies this, and Section 7 compares the identification structure with Millikan’s contact-EMF problem.

The estimating model has three effective parameters:

$$\boldsymbol{\theta} = (\mathcal{M}, \Omega_M, \Omega_\Lambda), \quad (17)$$

plus an intrinsic scatter σ_{int} . For a flat universe the constraint $\Omega_M + \Omega_\Lambda = 1$ reduces the shape parameters to the single free parameter Ω_M .

4 The Historical Data

4.1 The Low-Redshift Calibration Sample

Measuring $(\Omega_M, \Omega_\Lambda)$ requires supernovae spanning a wide enough redshift range that the *shape* of the $\mu(z)$ curve—which changes between different cosmological models—is statistically distinguishable. Both teams anchored their analyses with a common low-redshift ($z \lesssim 0.1$) training set: the Calán/Tololo Supernova Survey of Hamuy et al. [1996], which provided 29 carefully observed Type Ia supernovae in *BVI* bands. The SCP used 18 of them; the HZT drew its 34 low- z events more widely from the literature. Table 1 lists a representative subset.

At low redshifts $d_L \approx cz/H_0$ (Hubble’s law), so $\mu \approx 5 \log_{10}(cz/H_0 \cdot \text{Mpc}) + 25$. The shape parameters enter at order z , through the combination $q_0 = \Omega_M/2 - \Omega_\Lambda$, and Appendix B works out the coefficient. They are nonetheless unmeasurable from $z \lesssim 0.1$ data, because the leverage $\sum_i (z_i - \bar{z})^2$ over so narrow a range is too small: the low- z sample alone determines q_0 only to ± 1.1 . It constrains $\mathcal{M}$ and leaves the cosmology open.

4.2 The High-Redshift Samples

Both teams spent several years assembling high- z supernova samples using ground-based telescopes supplemented by Hubble Space Telescope photometry and Keck spectroscopy.

Table 1: Representative Type Ia supernovae from the Calán/Tololo low-redshift sample [Hamuy et al., 1996]. The effective B -band peak magnitude m_B^{eff} incorporates the light-curve shape correction of equation (5). These events serve as the “training data” in section 13: they calibrate the absolute magnitude M_B for use in the high- z sample.

Supernova	z	m_B^{eff} (mag)	σ (mag)
SN 1990O	0.030	16.33	0.07
SN 1992P	0.026	16.08	0.07
SN 1992ae	0.075	18.44	0.09
SN 1992al	0.014	14.60	0.06
SN 1992bc	0.020	15.18	0.06
SN 1992bh	0.045	17.42	0.08
SN 1992bl	0.043	17.29	0.08
SN 1992bo	0.018	15.95	0.06
SN 1993ag	0.050	17.68	0.09
SN 1993O	0.052	17.72	0.08

Perlmutter et al. (1999). The SCP’s “Fit C” sample comprises 18 high-quality high- z events; their full “Fit C+D” sample extends to 42 supernovae at $z = 0.18$ – 0.83 . Table 2 lists a representative subset, showing the effective apparent magnitudes after K -correction (accounting for the redshift of the SN Ia spectral energy distribution relative to the B -band filter) and stretch-factor standardisation.

Riess et al. (1998). The HZT’s primary dataset comprised 16 high- z supernovae ($z = 0.16$ – 0.97), supplemented by 34 low- z events compiled from the literature. The MLCS method was applied to multi-band ($BVRI$) light curves, providing simultaneous estimates of the shape parameter Δ , the colour excess $E(B - V)$ (dust reddening), and the peak magnitude. A subset is shown in Table 3.

4.3 Provenance of the Tabulated Data

Tables 1–3 are representative subsets transcribed from the published tables. I have not checked every entry against the originals, and no result in this paper is computed from them. The parameter estimates in Section 8 are the two teams’ own published values. The forecasts in Section 6 use the observed redshift distributions and quoted uncertainties, not the individual magnitudes.

The tables are still informative about scale. Referred to a flat Λ CDM model with

Table 2: Representative high-redshift Type Ia supernovae from the Supernova Cosmology Project [Perlmutter et al., 1999]. Effective B -band magnitudes m_B^{eff} after stretch-factor and K -correction. The photometric uncertainty σ includes contributions from photon noise, host-galaxy subtraction, K -correction uncertainty, and intrinsic scatter $\sigma_{\text{int}} \approx 0.15$ mag. Values are approximately reproduced from Table 1 of Perlmutter et al. [1999].

Supernova	z	m_B^{eff} (mag)	σ (mag)
SN 1994am	0.372	22.46	0.21
SN 1994H	0.374	22.17	0.15
SN 1994an	0.378	22.92	0.18
SN 1995az	0.450	22.79	0.20
SN 1994al	0.420	22.45	0.22
SN 1994G	0.425	22.55	0.17
SN 1994F	0.354	22.24	0.19
SN 1995ba	0.388	22.65	0.17
SN 1992bi	0.458	22.92	0.16
SN 1996cg	0.490	23.38	0.18
SN 1997ap	0.830	24.23	0.23

$H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the Calán/Tololo entries imply M_B between -18.5 and -19.5 , with a mean near -19.2 and a dispersion of about 0.25 mag. The SCP high- z entries imply a mean M_B near -19.1 . The difference between the two means is the acceleration signal. The difference is about 0.1 mag in this subset, against the 0.25 mag that the full samples deliver. A dozen events with 0.2 mag errors show no more.

4.4 The Standardisation Pipeline

Both teams corrected the raw observed magnitudes for four effects before entering them into the cosmological likelihood:

1. **K -correction.** At $z \sim 0.5$, the observer-frame I -band covers rest-frame B -band, while the observer-frame B -band covers rest-frame UV. A wavelength-dependent correction transforms the measured filter magnitude to the rest-frame B -band magnitude that anchors the low- z calibration. The SCP used templates of Kim et al. [1996]; the HZT used spectrophotometric templates from Nugent et al. [2002].
2. **Light-curve shape correction.** The SCP employed a *stretch factor* s (time-axis scaling): each light curve is remapped to the template by stretching the time axis

Table 3: Selected high-redshift supernovae from the High-Z Supernova Search Team [Riess et al., 1998]. Corrected distance moduli μ_{MLCS} are computed from the full MLCS fit to the multi-band light curves and incorporate corrections for light-curve shape (Δ) and dust reddening (A_B). The MLCS zero point is arbitrary and is absorbed into $\mathcal{M}$, so these moduli carry a common offset relative to any particular cosmology; only their differences matter. Values approximately reproduced from Riess et al. [1998], Table 5.

Supernova	z	μ_{MLCS}	σ_μ	Notes
SN 1997ce	0.440	42.53	0.33	“golden”
SN 1997cj	0.500	42.91	0.24	“golden”
SN 1997ck	0.970	44.30	0.51	no spectrum
SN 1995ao	0.300	41.08	0.36	
SN 1995ap	0.230	40.36	0.26	
SN 1996cf	0.570	43.16	0.42	
SN 1996bj	0.574	43.67	0.26	

by s , then the peak magnitude is corrected by $\alpha_{\text{SCP}}(s - 1)$ with $\alpha_{\text{SCP}} \approx 1.35$ mag. The HZT’s MLCS method fit multi-band templates parameterised by a single luminosity offset Δ ; the correction is $\eta \Delta$ with $\eta \approx 0.91$ mag per unit Δ .

3. **Colour/dust correction.** Both teams corrected for dust reddening in the host galaxy using the measured $B - V$ colour excess at peak light: $A_B = R_B E(B - V)$ with $R_B \approx 4.1$.
4. **Peculiar-velocity uncertainty.** Each SN at $z < 0.1$ contributes a velocity uncertainty $\sigma_v \approx 200\text{--}300$ km/s from large-scale structure motions, adding an effective distance- modulus error $\sigma_{v,i} = 5\sigma_v / (cz_i \ln 10)$ that diverges at low z . This is why the minimum useful redshift is $z \approx 0.01$.

After these corrections, the effective m_B^{eff} enters the cosmological likelihood as if it were the true distance modulus plus a Gaussian residual.

The total photometric uncertainty per supernova is typically

$$\sigma_{m,i}^2 = \sigma_{\text{photo},i}^2 + \sigma_{K,i}^2 + \sigma_{v,i}^2 + \sigma_{\text{int}}^2,$$

where σ_{int} is the residual dispersion after standardisation, taken as 0.15 mag throughout (Section 2.2).

5 Weighted Least Squares and Maximum Likelihood Estimation

5.1 The Likelihood Function

Given the measurement model (16), in which each observation $m_{B,i}$ is drawn from a normal distribution centred on $\mathcal{M} + 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda)$, the likelihood for a sample of $n = N_{\text{low}} + N_{\text{high}}$ supernovae is

$$L(\boldsymbol{\theta}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma_i} \exp\left(-\frac{[m_{B,i} - \mathcal{M} - 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda)]^2}{2\sigma_i^2}\right), \quad (18)$$

where the total uncertainty σ_i (including intrinsic scatter) was discussed in Section 4. The log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2} \ln(2\pi) - \sum_{i=1}^n \ln \sigma_i - \underbrace{\frac{1}{2} \sum_{i=1}^n \frac{[m_{B,i} - \mathcal{M} - 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda)]^2}{\sigma_i^2}}_{\chi^2(\boldsymbol{\theta})/2}. \quad (19)$$

Maximising (19) over $\boldsymbol{\theta}$ minimises $\chi^2(\boldsymbol{\theta})$.

5.2 MLE Equals WLS Under Gaussian Errors

For fixed σ_i (i.e., ignoring the dependence of σ_{int} on $\boldsymbol{\theta}$), maximising (19) over $(\mathcal{M}, \Omega_M, \Omega_\Lambda)$ requires minimising

$$\chi^2(\mathcal{M}, \Omega_M, \Omega_\Lambda) = \sum_{i=1}^n \frac{[m_{B,i} - \mathcal{M} - 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda)]^2}{\sigma_i^2}. \quad (20)$$

Unlike the photoelectric case—where OLS had a closed-form solution—minimising (20) over $(\Omega_M, \Omega_\Lambda)$ requires numerical optimisation because $\tilde{d}_L(z; \Omega_M, \Omega_\Lambda)$ involves a numerical integral rather than an algebraic formula. Both groups used grid search over a fine $(\Omega_M, \Omega_\Lambda)$ lattice, evaluating χ^2 at each grid point via numerical quadrature. The estimator is still the minimiser:

$$\boxed{(\hat{\mathcal{M}}_{\text{MLE}}, \hat{\Omega}_{M,\text{MLE}}, \hat{\Omega}_{\Lambda,\text{MLE}}) = \arg \min_{(\mathcal{M}, \Omega_M, \Omega_\Lambda)} \chi^2(\mathcal{M}, \Omega_M, \Omega_\Lambda).} \quad (21)$$

5.3 Profiling Out the Nuisance Parameter

$\mathcal{M}$ can be eliminated analytically at any fixed $(\Omega_M, \Omega_\Lambda)$, because it enters (20) linearly:

$$\hat{\mathcal{M}} = \frac{\sum_i \frac{m_{B,i} - 5 \log_{10} \tilde{d}_L(z_i)}{\sigma_i^2}}{\sum_i \frac{1}{\sigma_i^2}}. \quad (22)$$

Substituting (22) into χ^2 gives the *profile chi-squared*

$$\tilde{\chi}^2(\Omega_M, \Omega_\Lambda) \equiv \chi^2(\hat{\mathcal{M}}, \Omega_M, \Omega_\Lambda), \quad (23)$$

which depends only on the two shape parameters and can be minimised on a two-dimensional grid. In practice both groups also marginalised or profiled over intrinsic scatter σ_{int} .

5.4 Predicted Distance Moduli for Representative Cosmological Models

Table 4 computes the theoretical distance modulus $\mu^{\text{th}}(z; \Omega_M, \Omega_\Lambda, H_0)$ for four cosmological models at representative redshifts, using $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The model predictions diverge most strongly at $z \approx 0.3\text{--}0.8$.

At $z \approx 0.5$ flat Λ CDM predicts supernovae 0.22 mag fainter than the open matter-only model and 0.40 mag fainter than Einstein–de Sitter. The total per-supernova uncertainty at that redshift was about 0.22 mag, of which 0.15 is intrinsic dispersion after standardisation and the rest photometric and K -correction error. Each event therefore carries about one standard deviation of signal against the open model.

Naive stacking would suggest that 40 such events settle the question at $\sqrt{40} \approx 6\sigma$. Both teams claimed about 3σ . The gap is the subject of Section 6. Stacking assumes that the quantity of interest is a single mean offset at fixed Ω_M . The teams estimated a two-dimensional parameter. Marginalising over Ω_M along the degeneracy direction

Table 4: Theoretical distance modulus $\mu = 5 \log_{10}(d_L/\text{Mpc}) + 25$ for $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ under three cosmological models, computed by numerical integration of equation (7). $\Delta\mu_\Lambda$ is the excess distance modulus of flat ΛCDM over the open matter-only model, and $\Delta\mu_{\text{EdS}}$ its excess over Einstein–de Sitter. Positive values mean fainter supernovae.

z	μ^{th} (mag)			$\Delta\mu_\Lambda$ (Λ –open)	$\Delta\mu_{\text{EdS}}$ (Λ –EdS)
	Flat ΛCDM (0.3, 0.7)	EdS (1.0, 0.0)	Open (0.3, 0.0)		
0.10	38.32	38.21	38.25	+0.07	+0.11
0.25	40.50	40.27	40.36	+0.14	+0.24
0.50	42.26	41.86	42.05	+0.22	+0.40
0.75	43.33	42.82	43.08	+0.25	+0.51
1.00	44.10	43.50	43.84	+0.26	+0.60

costs most of the apparent significance. Systematic errors, which do not fall as $1/\sqrt{n}$, consumed more.

The excess faintness found in both analyses, $\Delta\mu_{\text{obs}} \approx +0.25$ mag at $z \approx 0.5$ (Section 8), exceeds the ΛCDM prediction over the open model and falls well short of the prediction over Einstein–de Sitter.

6 Fisher Information and Confidence Ellipses

6.1 The Fisher Information Matrix

The precision with which $(\Omega_M, \Omega_\Lambda)$ can be estimated is governed by the Fisher information matrix $\mathcal{I}(\boldsymbol{\theta})$. For the model (16) with known σ_i :

$$\mathcal{I}_{jk}(\boldsymbol{\theta}) = \sum_{i=1}^n \frac{1}{\sigma_i^2} \frac{\partial \mu^{\text{th}}(z_i; \boldsymbol{\theta})}{\partial \theta_j} \frac{\partial \mu^{\text{th}}(z_i; \boldsymbol{\theta})}{\partial \theta_k}, \quad (24)$$

where the sum runs over all supernovae. The Cramér–Rao bound states that any unbiased estimator $\tilde{\boldsymbol{\theta}}$ satisfies $\text{Var}(\tilde{\theta}_j) \geq [\mathcal{I}^{-1}]_{jj}$.

For the shape parameters $(\Omega_M, \Omega_\Lambda)$ the relevant derivatives are

$$\frac{\partial \mu}{\partial \Omega_M} = \frac{5}{\ln 10} \frac{1}{\tilde{d}_L} \frac{\partial \tilde{d}_L}{\partial \Omega_M}, \quad \frac{\partial \mu}{\partial \Omega_\Lambda} = \frac{5}{\ln 10} \frac{1}{\tilde{d}_L} \frac{\partial \tilde{d}_L}{\partial \Omega_\Lambda}. \quad (25)$$

These are computed numerically. Ω_M slows the expansion and reduces μ . Ω_Λ accelerates the expansion and raises μ . The two derivatives have opposite signs and comparable magnitudes over the whole observed range, so the parameters are nearly degenerate. At $z = 0.5$ and the best-fit parameters,

$$\frac{\partial\mu/\partial\Omega_M}{\partial\mu/\partial\Omega_\Lambda} \approx -1.1, \quad (26)$$

falling to -0.96 at $z = 0.4$ and rising to -1.24 at $z = 0.6$. Weighted across an SCP-like sample the effective ratio is -1.5 . Raising Ω_M by one unit and Ω_Λ by about 1.5 units leaves $\mu(z)$ nearly unchanged.

The Fisher matrix accordingly has one large and one small eigenvalue in the $(\Omega_M, \Omega_\Lambda)$ block. The well-measured combination is

$$0.8\Omega_M - 0.6\Omega_\Lambda, \quad (27)$$

which [Perlmutter et al. \[1999\]](#) report as -0.2 ± 0.1 . The orthogonal combination $0.6\Omega_M + 0.8\Omega_\Lambda$ is nearly unconstrained. The confidence ellipses are narrow along (27) and elongated along the orthogonal direction. [Goobar and Perlmutter \[1995\]](#) derived this geometry before the data existed and designed the survey around it.

6.2 A Fisher Forecast for the 1998 Designs

Table 5 evaluates (24) for four designs. Each uses 18 low- z supernovae spread over $z \in [0.02, 0.10]$ with $\sigma = 0.17$ mag, the intrinsic 0.15 combined with the peculiar-velocity term of Section 4, and differs only in the high- z sample. I profile over $\mathcal{M}$ and report standard errors on the density parameters. The calculation uses the redshift distributions and quoted uncertainties, not the individual magnitudes.

The individual density parameters are not measured. The SCP-like design gives $\text{SE}(\Omega_M) = 0.44$ and $\text{SE}(\Omega_\Lambda) = 0.65$, so neither parameter is determined to better than a factor of two. The published ± 0.09 on Ω_M is not a marginal standard error. It measures precision under the flatness constraint $\Omega_M + \Omega_\Lambda = 1$, which the same calculation gives as 0.070.

One combination is measured well. The forecast standard error on $0.8\Omega_M - 0.6\Omega_\Lambda$ is 0.10, matching the ± 0.1 that [Perlmutter et al. \[1999\]](#) report. The agreement of a

Table 5: Fisher-information standard errors for four high- z designs, each paired with 18 low- z supernovae at $\sigma = 0.17$ mag and profiled over $\mathcal{M}$. High- z errors are $\sigma = 0.22$ mag except the HZT-like row, which uses 0.25 mag. The last column is the standard error on the combination (27).

High- z design	n	z range	$\text{SE}(\Omega_M)$	$\text{SE}(\Omega_\Lambda)$	$\text{SE}(0.8\Omega_M - 0.6\Omega_\Lambda)$
SCP-like	42	0.18–0.83	0.44	0.65	0.10
Truncated	42	0.18–0.62	0.81	0.98	0.13
Bunched	42	0.35–0.55	1.30	1.50	0.12
HZT-like	16	0.16–0.62	1.32	1.56	0.19

forecast built from redshifts and error bars alone with the published error bar confirms that the design fixed the 1998 constraint, and the details of the photometry did little to it.

Redshift range buys precision on the individual parameters and almost none on the combination. Truncating the sample at $z = 0.62$ nearly doubles $\text{SE}(\Omega_M)$ and widens the combination only from 0.10 to 0.13. Bunching all 42 events at $z \approx 0.45$ triples $\text{SE}(\Omega_M)$ and leaves the combination at 0.12. The events beyond $z = 0.6$ rotate the ellipse. The rotation separates Ω_M from Ω_Λ .

6.3 The Value of Wide Redshift Coverage

Goobar and Perlmutter [1995] showed that separating Ω_M from Ω_Λ requires supernovae at two well-separated redshifts. A sample restricted to $z \approx 0.5$ traces a near-degenerate path in the $(\Omega_M, \Omega_\Lambda)$ plane. Adding $z \approx 0.8$ –1.0 events rotates the ellipse, as the second and third rows of Table 5 show.

The SCP’s highest-redshift event, SN 1997ap at $z = 0.83$, carried a large share of that rotation. The HZT’s highest, SN 1997ck at $z = 0.97$, carried more still, and lacked a confirming spectrum; Riess et al. [1998] reported their conclusions with and without it.

6.4 Numerical Illustration of the Confidence Ellipses

Perlmutter et al. [1999] reported their confidence contours directly as $\Delta\chi^2 = \tilde{\chi}^2 - \tilde{\chi}_{\min}^2$ contours. The 68% and 95% confidence ellipses for one free parameter correspond to $\Delta\chi^2 = 1$ and 4; for two parameters, to $\Delta\chi^2 = 2.30$ and 5.99. In the published

confidence plots (Figure 2 of [Perlmutter et al. 1999](#); Figure 7 of [Riess et al. 1998](#)) the best-fit region lies in the quadrant $\Omega_\Lambda > 0$.

The ellipse is elongated, and the line $\Omega_\Lambda = 0$ runs nearly along the elongation, so the distance between them depends on how it is measured. Measured by the marginal standard error on Ω_Λ alone, the SCP-like forecast of Table 5 puts $\Omega_\Lambda = 0.72$ only 1.1σ from zero. The physical restriction $\Omega_M \geq 0$ does the remaining work. Setting $\Omega_\Lambda = 0$ and minimising over $\Omega_M \geq 0$ raises χ^2 by 3.8 in the same forecast, or 2.0σ . The published 3σ figures come from the data and the full error budget, not from this forecast. The forecast locates their strength. No non-negative Ω_M reproduces the observed curvature with $\Omega_\Lambda = 0$, and that is what excludes a decelerating universe. The size of $\hat{\Omega}_\Lambda$ on its own does not.

Under flatness the picture changes. The same forecast gives $\text{SE}(\Omega_M) = 0.070$, so Einstein–de Sitter ($\Omega_M = 1$) sits 10σ from the best fit. The strength of the 1998 result rests on the joint use of the supernova data and a prior—either $\Omega_M \geq 0$, or flatness from the CMB. Section 12 returns to what that dependence does and does not license.

7 The $\mathcal{M}$ Identification Problem

7.1 The Contact-EMF Analogy

In Millikan’s photoelectric regression the contact electromotive force v_c between the photoemitting metal and the collector entered the raw readings as a pure intercept shift:

$$V_{0i}^{\text{obs}} = \underbrace{-W/e - v_c}_\alpha + (h/e) \nu_i + \varepsilon_i.$$

The gradient of the V_0 – ν relation identifies the slope h/e . Its level does not. Millikan measured v_c continuously with a Kelvin double-balance and found it constant over a run, so it shifted all observations equally and left $\hat{\beta}$ unbiased.

In the supernova problem $\mathcal{M} = M_B + 5 \log_{10}(c/H_0 \text{ Mpc}) + 25$ plays the role of v_c . The curvature of the m_B – z relation carries the cosmology. Its level does not. Neither M_B nor H_0 can be determined from the Hubble diagram alone, because a change in H_0 can be offset exactly by a shift in M_B . The shape parameters escape that degeneracy.

The parallel with Millikan ends at that point, and the difference is the harder half of the

problem. Millikan’s equation was linear and its slope was a single number, so removing v_c left h/e fully identified. The Friedmann model has two shape parameters, and removing $\mathcal{M}$ leaves them tangled with each other. A second degeneracy therefore survives the first: the supernovae fix $0.8\Omega_M - 0.6\Omega_\Lambda$ and leave the orthogonal combination open (Section 6). The contact EMF was an apparatus nuisance that better instruments could remove. This one is a property of the geometry, and no supernova measurement removes it.

7.2 Constraining $\mathcal{M}$ from the Low- z Sample

Both teams’ solution was identical in spirit to Millikan’s: constrain the nuisance parameter from an independent source. At low z , $d_L \approx cz/H_0$, so $m_{B,i} \approx \mathcal{M} + 5\log_{10} z_i + C$ (a constant). The Calán/Tololo low- z sample calibrates $\mathcal{M}$ through Hubble’s law (1), with uncertainties from peculiar velocities averaged away over the 18 events the SCP used. Once $\mathcal{M}$ is anchored by the low- z data, the high- z data constrain $(\Omega_M, \Omega_\Lambda)$ through the curvature of $\mu(z)$.

7.3 Separate Identification of H_0 and M_B

Even with the low- z anchor, H_0 and M_B are individually identified only if M_B is calibrated by an external distance ladder. The primary constraint came from Cepheid variable distances measured with the Hubble Space Telescope Key Project, which established $H_0 = 72 \pm 8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (circa 1998). Given H_0 , the Calán/Tololo low- z sample then determines $M_B \approx -19.3 \text{ mag}$.

M_B , equivalently H_0 , is a nuisance parameter for the cosmological inference. Its uncertainty propagates into $\mathcal{M}$ but not into $(\Omega_M, \Omega_\Lambda)$. A 10% error in H_0 shifts the Hubble diagram by $\approx 0.22 \text{ mag}$ and leaves the best-fit Ω_M and Ω_Λ unchanged.

8 Results: The Deceleration Parameter and Cosmic Acceleration

8.1 Best-Fit Parameters

Table 6 summarises the principal parameter estimates from both groups, using their primary standardisation methods.

Table 6: Cosmological parameter estimates from the two Nobel-Prize groups. Every published row imposes flatness, $\Omega_M + \Omega_\Lambda = 1$; without it neither parameter is separately determined, as the last row records. Statistical uncertainties only. The deceleration parameter is $q_0 = \Omega_M/2 - \Omega_\Lambda$.

Analysis	$\hat{\Omega}_M$	$\hat{\Omega}_\Lambda$	$\hat{q}_0$	Ref.
Perlmutter et al. (SCP), flat	0.28 ± 0.09	0.72 ± 0.09	-0.58	Perlmutter et al. [1999]
Riess et al. (HZT), MLCS, flat	0.24 ± 0.10	0.76 ± 0.10	-0.64	Riess et al. [1998]
Riess et al. (HZT), template, flat	0.20 ± 0.10	0.80 ± 0.10	-0.70	Riess et al. [1998]
<i>Standard errors without the flatness constraint (forecast):</i> $\text{SE}(\Omega_M) = 0.44$, $\text{SE}(\Omega_\Lambda) = 0.65$ (Table 5)				
Planck 2018 (CMB+BAO, flat)	0.315 ± 0.007	0.685 ± 0.007	-0.528	Planck Collaboration et al. [2020]

Both groups found $\hat{\Omega}_\Lambda$ between 0.72 and 0.80 from independent datasets and different standardisation pipelines, each about one standard error wide. Section 9 compares them. Every published estimate in the table assumes flatness. The supernova data alone do not deliver these numbers.

8.2 The Deceleration Parameter

From the second Friedmann equation,

$$q_0 \equiv -\frac{\ddot{a} a}{\dot{a}^2} \Big|_{t=t_0} = \frac{\Omega_M}{2} - \Omega_\Lambda. \quad (28)$$

For the best-fit flat model ($\hat{\Omega}_M = 0.28$, $\hat{\Omega}_\Lambda = 0.72$):

$$\hat{q}_0 = \frac{0.28}{2} - 0.72 = 0.14 - 0.72 = -\mathbf{0.58}. \quad (29)$$

The universe is accelerating. The transition from deceleration to acceleration occurred at the redshift z_t satisfying $\Omega_M(1 + z_t)^3 = 2\Omega_\Lambda$, which gives

$$z_t = \left(\frac{2\Omega_\Lambda}{\Omega_M}\right)^{1/3} - 1 \approx \left(\frac{1.44}{0.28}\right)^{1/3} - 1 \approx (5.14)^{1/3} - 1 \approx 0.73. \quad (30)$$

At $z \lesssim 0.73$, dark energy dominates and expansion accelerates; at $z \gtrsim 0.73$, matter dominates and expansion decelerates. Riess et al. [2004] later found supernovae at $z > 1$ showing the predicted past deceleration. Section 11 treats that observation and the transition redshift they report.

8.3 The Residual Hubble Diagram

The residual Hubble diagram is the difference between observed distance moduli and those predicted by the best-fit matter-only open model. Define

$$\Delta\mu_{\text{obs},i} \equiv \mu_{\text{obs},i} - \mu^{\text{th}}(z_i; \Omega_M = 0.3, \Omega_\Lambda = 0). \quad (31)$$

For the best-fit flat Λ CDM model, $\Delta\mu^{\text{th}}(z = 0.5) \approx +0.21$ mag (Table 4).

From both datasets, the weighted mean residual for high- z supernovae is

$$\overline{\Delta\mu}_{\text{obs}}(z \approx 0.3\text{--}0.7) \approx +0.25 \pm 0.06 \text{ mag}. \quad (32)$$

Taken as a single mean offset, the excess sits 4σ from $\Delta\mu = 0$, and it matches the flat Λ CDM prediction to within 1σ . That 4σ overstates the evidence for acceleration, for the reason given in Section 6: the comparison holds Ω_M fixed at 0.3, and the published significances marginalise over it.

The supernovae are fainter, hence further away, than expected. The universe expanded more between then and now than a decelerating universe would have.

8.4 Statistical Significance

For Perlmutter et al. (1999):

- $\chi^2_{\text{min}}/\nu \approx 1.03$ for the best-fit flat model (42 SNe, 2 free parameters after profiling $\mathcal{M}$, so $\nu = 39$ degrees of freedom).
- The no- Λ hypothesis ($\Omega_\Lambda = 0$) gives $\Delta\chi^2 \approx 10.8$ (1 d.o.f.), corresponding to $p < 0.001$, i.e., rejected at 3.3σ .
- Profiling over both parameters for the constraint $q_0 \geq 0$ (decelerating universe) gives rejection at $> 99.9\%$ confidence.

For Riess et al. (1998), the MLCS method gave:

- $q_0 < 0$ at 3.0σ using MLCS and 2.8σ using template fitting. These figures come from a secondary source and should be checked against [Riess et al. \[1998\]](#).
- The two methods agreed across independent analysis pipelines, so the result is not an artifact of the MLCS standardisation.

9 Comparing the Two Teams

Table 7 summarises the differences and agreements between the two groups.

Table 7: Comparison of the two groups’ datasets, methods, and principal results. Both teams reached the same conclusion from independent data and different analysis pipelines.

Feature	High-Z Team [Riess et al., 1998]	Supernova Cosmology Project [Perlmutter et al., 1999]
High- z sample size	16 SNe at $z = 0.16$ – 0.62	42 SNe at $z = 0.18$ – 0.83
Low- z sample	34 SNe (literature)	18 SNe (Calán/Tololo)
Standardisation	MLCS (multi-colour light curve templates; Δ parameter)	Stretch factor (s ; time-axis scaling of single-colour template)
Colour correction	$E(B-V)$ from multi-colour light curve fit	$E(B-V)$ from $B-V$ at peak
K -correction	Nugent et al. [2002] spectral templates	Kim et al. [1996] colour-matching
Intrinsic scatter	$\sigma_{\text{int}} = 0.14$ mag	$\sigma_{\text{int}} = 0.13$ mag
Primary result (flat)	$\hat{\Omega}_M = 0.24 \pm 0.10$	$\hat{\Omega}_M = 0.28 \pm 0.09$
q_0 significance	$q_0 < 0$ at 3.0σ (MLCS)	$q_0 < 0$ at $> 3\sigma$

The teams used different telescopes, observing strategies, standardisation algorithms, and statistical analysis packages. Their best-fit $\hat{\Omega}_M$, 0.28 and 0.24, differ by less than half a standard error.

The agreement tests less than it appears to in one respect and more in another. Both numbers assume flatness, and under flatness the supernovae measure a single parameter, so two teams agreeing on one number with ± 0.10 error bars is a weak test. Against that, the two pipelines shared almost nothing. MLCS and stretch parameterise the light curve differently, the K -corrections came from different spectral templates, the

colour corrections used different estimators, and the high- z samples overlapped barely at all. A systematic error in any one of those steps would have to have been reproduced by an independent implementation to survive. Replication across methods tests the standardisation, not the statistics.

10 Epistemic Scepticism and Robustness Tests

10.1 The Initial Reaction

Millikan’s 1916 confirmation of Einstein’s photoelectric equation was greeted with scepticism about the physical interpretation, even as the equation itself was accepted. The supernova result of 1998–1999 provoked a structurally similar reaction: many physicists accepted the statistical result but questioned whether it truly implied $\Omega_\Lambda > 0$, proposing instead that systematic errors mimicked the acceleration signal.

The principal alternative explanations advanced were:

1. **Dust dimming.** If a smooth screen of “grey” (wavelength- independent) dust occupied intergalactic space, it would absorb SN light at all redshifts preferentially at high z , making distant SNe appear fainter without the colour reddening that normally betrays dust. Both teams addressed this by checking the colour residuals of their SNe: if grey dust were responsible, the $B - V$ colour excess would scale with the magnitude offset, but no such correlation was found at the level required to explain the signal.
2. **Luminosity evolution.** If Type Ia supernovae at $z \sim 0.5$ were intrinsically ≈ 0.25 mag fainter than their low- z counterparts (due to differences in progenitor age, composition, or environment), the signal could be entirely spurious. The teams tested this by comparing the Phillips Δm_{15} distributions and colour distributions of high- z and low- z SNe; no systematic difference was found. [Riess et al. \[2004\]](#) later supplied the decisive test with supernovae at $z > 1$. Section 11 gives it.
3. **Malmquist bias.** At high z , magnitude-limited surveys preferentially detect intrinsically bright events, biasing $\hat{m}_B$ brighter. Both teams corrected for this with detailed simulations; the bias was found to be negligible compared to the acceleration signal for their survey depths.

The acceleration signal survived all of these checks.

10.2 CMB and Large-Scale Structure Evidence

Independent corroboration came from the cosmic microwave background. The BOOMERANG balloon experiment [2000] detected the first acoustic peak of the CMB temperature power spectrum and confirmed that the total energy density satisfies $\Omega_{\text{tot}} \approx 1$ (flat universe). Combined with Hubble-constant and large-scale-structure constraints that independently fix $\Omega_M \approx 0.3$, this implies $\Omega_\Lambda \approx 0.7$, consistent with the supernova result within the errors of both, which were then of order 10%.

The corroboration matters more than the agreement. The supernova constraint runs along one direction of the $(\Omega_M, \Omega_\Lambda)$ plane and the CMB constraint along a nearly orthogonal one, so the two together determine both parameters where neither does alone. The flatness that the published supernova results assume is exactly what the CMB supplies.

By 2020 Planck [Planck Collaboration et al., 2020] gave $\Omega_\Lambda = 0.685 \pm 0.007$, about 98 standard errors from zero, and confirmed the 1998–1999 estimates to sub-percent precision.

11 The Prediction at $z > 1$

The 1998 fits estimated two parameters from the data they fitted. The test described here estimated nothing.

Take the flat Λ CDM model with $\Omega_M = 0.28$ and $\Omega_\Lambda = 0.72$, fitted to supernovae at $z < 0.85$, and ask what it says about $z = 1.5$. Referred to Einstein–de Sitter, it predicts an excess faintness of +0.41 mag at $z = 0.5$ and only +0.74 mag at $z = 1.5$. The curve flattens, because dark energy dominates late and matter dominates early. Now take the leading systematic explanation of the same 1998 data. Grey dust and luminosity evolution both dim distant supernovae by amounts that grow with distance. Calibrated to reproduce the observed +0.41 mag at $z = 0.5$, a dimming proportional to z predicts +1.24 mag at $z = 1.5$ and keeps climbing.

The two models separate by half a magnitude at $z = 1.5$, two to three per-supernova standard deviations, which a handful of events settles. Referred instead to an empty universe the prediction is sharper still. Λ CDM puts supernovae beyond $z \approx 1.4$ *brighter* than that benchmark, the signature of the deceleration that preceded the dark-energy

Table 8: Excess faintness relative to Einstein–de Sitter under the flat Λ CDM model fitted in 1998, and under a dimming proportional to z calibrated to match it at $z = 0.5$. The last column is the difference the two models predict, in magnitudes. Per-supernova errors at these redshifts were 0.2–0.3 mag.

z	Λ CDM	Dimming $\propto z$	Difference
0.5	+0.41	+0.41	0.00
1.0	+0.62	+0.83	−0.20
1.5	+0.74	+1.24	−0.50

era. No model in which distant supernovae are dimmed by dust or by evolution produces a change of sign.

[Riess et al. \[2004\]](#) collected the supernovae at $z > 1$ with the Hubble Space Telescope and found the turnover. The 1998 parameters had no freedom left to accommodate the result. That observation is the out-of-sample test of the model. The 3σ of 1998 measured parameters within it.

One clarification about the transition redshift. The flat model fitted in 1998 implies $z_t = (2\Omega_\Lambda/\Omega_M)^{1/3} - 1 = 0.73$, and the Planck parameters imply 0.63. The figure $z_t \approx 0.46$ that [Riess et al. \[2004\]](#) report comes from a kinematic fit in which $q(z)$ is linear in z , not from Λ CDM, and the two numbers are not comparable. The qualitative prediction—that deceleration preceded acceleration, and that the turnover falls near $z \sim 0.5$ – 0.8 —is what the observation confirmed.

12 What the Supernovae Could Not Settle

The robustness tests of Section 10 answered questions about the data. A separate question concerns the model, and the supernovae do not answer it.

Type Ia supernovae measure one function, the luminosity distance $d_L(z)$. Every quantity in this paper is a functional of that one curve. Two theories that deliver the same $d_L(z)$ over $z \in [0, 1]$ are observationally identical to these data, whatever else they assert. Three classes of theory do so. A cosmological constant with $w = -1$ is one. A rolling scalar field with $w(z)$ within a few percent of -1 over the observed range is a second, and Appendix C records that current data bound w only to about 0.03. Modified theories of gravity that reproduce the same expansion history without any

dark-energy fluid are a third. The 1998 data select the functional form of $d_L(z)$. They do not select the physics that generates it.

The inference to acceleration itself rests on a prior. Section 6 showed that the supernovae measure $0.8\Omega_M - 0.6\Omega_\Lambda$ to ± 0.1 and the orthogonal combination hardly at all. Ruling out $\Omega_\Lambda = 0$ requires either $\Omega_M \geq 0$, which is a physical restriction rather than a measurement, or flatness, which the CMB supplies. Neither prior is controversial. Both are external to the supernova data, and the published error bars do not survive their removal.

Distinguishing Λ from a dynamical field requires a different observable. [Riess et al. \[2004\]](#) and the later surveys pursued one of them, the effect of w on $d_L(z)$ at higher redshift. The growth rate of large-scale structure supplies the other. Modified gravity and dark energy predict different growth rates even when they produce identical expansion histories, so growth separates them where distances cannot. Theory names the discriminating observable. No one would measure growth rates to test dark energy without a theory predicting that the two differ.

The mechanism remains unexplained. Quantum field theory predicts a vacuum energy density some 10^{120} times the observed Λ [[Weinberg, 1989](#)]. Riess, Schmidt, and Perlmutter measured the curvature of the Hubble diagram to a few percent. The discrepancy between that measurement and the theory of what it measures is the largest in physics.

13 The Physicists as Machine Learners

13.1 Overview

The vocabulary of modern machine learning—supervised learning, hypothesis class, loss function, inductive bias, feature engineering, transfer learning—did not circulate in the astrophysics community of the 1990s. The two teams’ reasoning maps onto it as Einstein’s 1905 reasoning does. [Sargent \[2025\]](#) argues that the mapping generally holds, and that the techniques now gathered under artificial intelligence have long histories inside the sciences that used them. This section states the correspondence for the supernova case.

13.2 The Hypothesis Class

The Friedmann equations and the definition of luminosity distance fix the hypothesis class for the supernova problem:

$$\mathcal{H} = \{m_B \mapsto \mathcal{M} + 5 \log_{10} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda) \mid \mathcal{M} \in \mathbb{R}, \Omega_M \geq 0, \Omega_\Lambda \in \mathbb{R}\}. \quad (33)$$

This is a three-parameter family of smooth curves on the (z, m_B) plane. Without the Friedmann equations the space of possible $m_B(z)$ functions is infinite-dimensional; with them it collapses to the three real numbers $(\mathcal{M}, \Omega_M, \Omega_\Lambda)$.

The theoretical restriction to $\mathcal{H}$ was not arbitrary. It follows deductively from three independent bodies of physics:

1. General relativity (the geodesic equation in an expanding spacetime);
2. The assumption of spatial homogeneity and isotropy (the cosmological principle);
3. The assumption that pressure-free matter obeys $\rho_M \propto a^{-3}$ and the cosmological constant is truly constant.

Each assumption had been tested independently long before the supernova search. The Friedmann hypothesis class was well-motivated, not curve-fitted.

13.3 Feature Engineering: The Phillips Width–Luminosity Relation

Raw photometry of Type Ia supernovae involves hundreds of measurements at different epochs and wavelengths. Phillips [1993] found that the post-peak decline rate Δm_{15} predicts the peak luminosity, which reduces that high-dimensional representation to a one-dimensional correction. The correction cut the dispersion from 0.4 to 0.15 mag, a factor of 2.7 in the noise and a factor of 7 in the number of supernovae needed for a given precision.

Against the open matter-only model the signal at $z \approx 0.5$ is 0.22 mag (Table 4). Detecting a 0.22 mag offset at 3σ takes about 30 supernovae at $\sigma = 0.4$ mag and about 4 at $\sigma = 0.15$ mag, before any allowance for the parameter degeneracy of Section 6 or for systematics. Both teams had roughly the number of events that the pre-Phillips dispersion would have required merely to see the effect, and neither could have separated

it from Ω_M .

A data-driven learner given raw light curves would have to extract the Phillips representation from the data alone. Phillips did exactly that.

13.4 The Inductive Half of the Discovery

The account so far credits theory with the work. General relativity fixed the functional form, three parameters remained, and fifty supernovae sufficed where a theory-free learner would have needed thousands. Half of the discovery does not fit that account.

Δm_{15} came from data. No theory of Type Ia explosions predicted that the post-peak decline rate tracks peak luminosity, and Phillips [1993] did not derive the relation. He plotted decline rate against absolute magnitude for nine nearby supernovae and fitted a line. The relation has since been partly rationalised—more ^{56}Ni yields both a brighter peak and a more opaque, slower-cooling ejecta—and it has not been deduced. Thirty years on, the width–luminosity relation remains an empirical regularity whose calibration is refitted with every new survey.

The progenitor physics behind it is unsettled in a way that matters here. Section 2.2 records that the Chandrasekhar-mass account no longer commands consensus. A theorist in 1993 with the correct progenitor model in hand could not have written down equation (5), and a theorist today still cannot.

The supernova discovery was therefore a hybrid. A deductively specified hypothesis class was bolted to an inductively discovered feature, and neither half would have sufficed. Without Friedmann there is no $d_L(z)$ to fit and no meaning for the fitted numbers. Without Phillips the scatter is twice the signal and no amount of theory recovers it. The one step in the episode where a data-driven procedure did the indispensable work was the one step where no theory was available to do it instead.

The pattern has continued. Modern supernova cosmology runs on SALT2 and SALT3, light-curve models trained on data rather than derived from explosion physics, and their training sets grow with each survey. Feature extraction in this field has become more machine-learned over time. The hypothesis class has stayed Friedmann’s. The division of labour that Section 14 describes is not a claim that induction fails; it is a claim about which part of the problem each method can reach.

13.5 Training Data and Generalisation

The *training data* for both analyses consisted of the low- z Calán/Tololo supernovae [Hamuy et al., 1996]: the ≈ 29 events at $z = 0.01$ – 0.10 that calibrated the nuisance parameter $\mathcal{M}$ (and, via H_0 , the absolute magnitude M_B).

The high- z events at $z = 0.3$ – 0.8 carried most of the cosmological information. They were not held out. Both $(\Omega_M, \Omega_\Lambda)$ and $\mathcal{M}$ were fitted to the low- z and high- z samples together, so neither set is test data for the cosmology. Section 11 describes the observation that was.

The standardisation did transfer out of sample. Both teams applied the MLCS or stretch correction learned from low- z supernovae, unchanged, to the high- z events.

- The HZT’s MLCS templates and the coefficient $\eta = 0.91$ mag were learned from low- z training data and then used to correct high- z events (out-of-sample prediction).
- The SCP’s stretch factor s and correction coefficient α_{SCP} were similarly trained on low- z data.

The implicit assumption—that Type Ia supernovae at $z \approx 0.5$ follow the same Phillips relation as nearby events—is a *transfer learning* assumption. Its validity was tested through the spectroscopic comparison of high- z and low- z SN Ia spectra: both teams found that the optical spectra of high- z events were, within measurement uncertainty, identical to those of low- z events of the same light-curve shape.

The cross- z universality of the shape–luminosity relation reduces the effective degrees of freedom and makes the inference tractable with a small sample.

13.6 Model Selection: Accelerating versus Decelerating Universes

At the time of both papers, four main cosmological models competed:

- **Einstein-de Sitter** ($\Omega_M = 1, \Omega_\Lambda = 0$): the theoretically favoured flat matter-only model.
- **Open matter-only** ($\Omega_M \approx 0.3, \Omega_\Lambda = 0$): consistent with galaxy cluster mass measurements.

- **Flat Λ CDM** ($\Omega_M \approx 0.3$, $\Omega_\Lambda \approx 0.7$): the dark-energy model.
- **Flat “QCDM” with dynamical dark energy** ($w \neq -1$): generalisations where Ω_Λ is replaced by a field with equation of state $w < -1/3$.

The supernova data, combined with the flat-universe CMB evidence, selected flat Λ CDM: the best-fit χ^2 under the winning model was lower by $\Delta\chi^2 > 10$ than under Einstein–de Sitter. Any modern criterion—AIC, BIC, Bayesian evidence—returns the same verdict.

The verdict is narrower than it sounds. The first three models on the list differ only in $(\Omega_M, \Omega_\Lambda)$, so the selection among them is parameter estimation within one hypothesis class. The fourth differs in w , and the supernova data of 1998 could not distinguish $w = -1$ from $w = -0.8$. The data rejected a decelerating expansion history. They left standing every theory that reproduces the observed $d_L(z)$, which Section 12 enumerates.

13.7 The Deductive-Inductive Loop

Table 9 translates the supernova discovery into the language of modern machine learning.

14 A Deep Neural Network on the Supernova Data

14.1 Two Hypothesis Classes Compared

The Friedmann-equations hypothesis class (33) has three free parameters after fixing σ_{int} . A deep neural network (DNN) for the regression $z \rightarrow m_B$ is a universal function approximator [Hornik et al., 1989]: a three-layer network with 16 ReLU neurons per layer has approximately

$$p = (1 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 1 + 1) = 593 \text{ parameters.}$$

For ReLU networks the pseudo-dimension grows like $p \log_2 p$, here about 5,500 [Vapnik, 1998]. Converting that into a required sample size admits no single answer. The uniform-convergence bound $O(d/\epsilon^2)$ gives $n \sim 10^7$ for $\epsilon = 1\%$ and is vacuous at that size. The practitioner’s rule $n \gtrsim 10p$ gives $n \gtrsim 6,000$, and I use it below. The SCP had 42 supernovae; the HZT had 50.

Capacity bounds of this kind are loose. Overparameterised networks generalise in

Table 9: The hypothetico-deductive loop in the supernova episode, translated into the language of modern machine learning.

Step	Astrophysical name	ML name
Propose Friedmann + Λ	Theoretical model	Model architecture
Derive $d_L(z; \Omega_M, \Omega_\Lambda)$	Prediction from GR	Hypothesis class
Phillips relation: use Δm_{15} , not raw m_B	Standardisation	Feature engineering
Calán/Tololo survey	Low- z calibration	Training data
High- z SN Ia searches	Discovery campaigns	Training-data extension
Minimise $\chi^2(\Omega_M, \Omega_\Lambda)$	Maximum likelihood	WLS training
$\Omega_\Lambda > 0$ at 3σ	Acceleration inferred	Parameter estimate
Deceleration seen at $z > 1$	Prediction confirmed	Out-of-sample test
Same result from SCP and HZT	Cross-group replication	Independent reimplementation

regimes where uniform-convergence theory says they cannot [Belkin et al., 2019], and nothing below rests on a claim that a network provably fails. The binding difficulty at $n = 42$ is identifiability rather than generalisation error.

14.2 Epoch I: Before Phillips (1993)

Before the width–luminosity relation was known, the Type Ia scatter was ≈ 0.4 mag. A DNN applied to raw (z, m_B) data would face an $n = O(20)$ dataset with $\sigma_{\text{noise}} \approx 0.4$ mag.

Underdetermination. With $n = 20$ and $p = 593$, the DNN is massively overparameterised. Gradient descent finds a minimum-norm interpolant that passes through all 20 training points but is unconstrained between them [Belkin et al., 2019].

Loss of the discrimination. The cosmological signal is ≈ 0.25 mag at $z \approx 0.5$ —smaller than the raw 0.4 mag scatter. Without the Phillips feature transformation, the DNN cannot separate signal from noise at all.

14.3 Epoch II: After Phillips (1993), Before Large Samples

The Calán/Tololo survey (1996) provides clean standardised photometry for 29 supernovae, of which the SCP used 18. Either number is far short of what a 593-parameter network requires.

No residual degrees of freedom. The Friedmann model leaves $n - 3 = 15$ residual degrees of freedom for $n = 18$ observations and three free parameters—enough to estimate $\hat{\sigma}_{\text{int}}$ and compute meaningful confidence intervals. The DNN has no such decomposition: with 593 parameters and 18 data points, every weight and bias is undertrained; there are no residual degrees of freedom to estimate noise variance or standard errors.

Cannot identify the degeneracy direction. The degeneracy direction $0.8\Omega_M - 0.6\Omega_\Lambda \approx \text{const}$ (Section 6) must be identified from the *curvature* of $\mu(z)$ over a wide redshift range. A DNN trained on 18 low- z events, all bunched in the near-linear regime $z < 0.1$, has no way to estimate this curvature: the functional form is undetermined. The Friedmann model, by contrast, predicts the curvature analytically from the density parameters, requiring only a small number of well-placed high- z observations to pin it down.

14.4 Epoch III: The 1998–1999 Discovery Datasets

The SCP’s 42-SN and HZT’s 50-SN samples are the operative training sets for the discovery.

Overfitting in the interpolation regime. With $n = 42$ and $p = 593$, the DNN still operates in the interpolation regime: it can memorise all 42 training points exactly, but its predictions between and beyond the training redshifts are unconstrained. In particular, the DNN has no mechanism to recover the physically meaningful separation $\hat{q}_0 = -0.58$ from the overfit weights.

No Cramér-Rao bound. The cosmological analysis produced confidence ellipses from $\Delta\chi^2$ contours; the Friedmann model reduces the fit to $n - 3 = 39$ degrees of freedom. A DNN has no analogue: because its residuals are zero by construction

(interpolation), there is no estimate of σ_{int} , no prediction interval, and no confidence surface over $(\Omega_M, \Omega_\Lambda)$.

No physical interpretation of weights. Even if the DNN happened to learn a curve close to $\mu(z; 0.28, 0.72)$, the weights of the network would not decompose into $(\hat{\Omega}_M, \hat{\Omega}_\Lambda)$. The connection between the learned function and the deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda$ would be opaque.

14.5 The Modern Era: Large Supernova Compilations

By the 2010s, supernova compilations reached several hundred events (e.g., [Conley et al. 2011](#), Joint Light-Curve Analysis). In this regime a DNN with appropriate architecture and regularisation could, in principle, learn the shape of $\mu(z)$ from the data alone and converge toward the Friedmann functional form. It would still not identify the function, extract the parameters, relate them to q_0 , or extrapolate past the training range:

- Identify that the learned function is $5 \log_{10} \tilde{d}_L(z)$ where $\tilde{d}_L$ satisfies a specific integral from general relativity.
- Extract $(\Omega_M, \Omega_\Lambda)$ from the network weights without supplying the Friedmann equations as an external constraint.
- Relate the curvature of $\mu(z)$ to the deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda$ and hence to the question of whether the universe is accelerating.
- Predict the $z > 1$ deceleration observed by [Riess et al. \[2004\]](#): the DNN trained on $z < 1$ data would extrapolate the accelerating trend, not the decelerating one, because it has no physical constraint requiring the cosmological transition.

Table 10 quantifies the contrast between the Friedmann model and a DNN at each epoch of the supernova program.

The photoelectric episode has the same structure. Einstein’s quantum hypothesis reduced an infinite-dimensional function-learning problem to a two-parameter line-fitting problem. Friedmann’s equations and the cosmological principle reduced the supernova distance-magnitude relation to a three-parameter curve, and Phillips reduced the scatter enough to make that curve measurable from ≈ 40 events. In both cases the theorists specified the smallest useful hypothesis class; the data then constrained the free parameters within it.

Table 10: Sample-complexity comparison between the three-parameter Friedmann model (pseudo-dimension = 3) and a three-layer DNN ($p = 593$ parameters, pseudo-dimension $\approx 5,500$). “Sufficient n ” uses the practitioner’s rule $n \gtrsim 10p \approx 6,000$ for the DNN, and four for the Friedmann model, one more than its free parameters. Sufficiency for the Friedmann model means enough residual degrees of freedom to fit and to assess the fit; it does not mean the parameters are identified, which requires high- z coverage as well.

Epoch	n available	Friedmann model	DNN (VC $\approx$ 5700)	DNN outcome
Pre-Phillips (before 1993)	~ 20 raw	Insufficient [†]	Insufficient	Cannot detect signal (noise > signal)
Calán/ Tololo 1996	29 (low- z)	Insufficient [†]	Insufficient	Cannot constrain curvature of $\mu(z)$
Discovery 1998	42–50	Sufficient	Insufficient ($\times 140$)	Overfits; no inference
JLA compilation 2011	~ 500	Sufficient	Insufficient ($\times 12$)	Partially learns shape; cannot extract q_0
Modern era 2020s	~ 2000	Sufficient	Insufficient ($\times 3$)	Recovers $\mu(z)$ shape; cannot identify q_0

[†] Without high- z data, $(\Omega_M, \Omega_\Lambda)$ are not identified: only $\mathcal{M}$ can be constrained from low- z observations.

15 Conclusions

The supernova discovery of cosmic acceleration is, at its mathematical core, a three-parameter weighted nonlinear regression. The free parameters are $\mathcal{M}$ (a nuisance combination of absolute magnitude and Hubble constant), Ω_M (matter density), and Ω_Λ (cosmological constant / dark energy density). The measurement equation

$$m_{B,i} = \mathcal{M} + 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda) + \varepsilon_i$$

is derived directly from general relativity; the Friedmann equations are the physical hypothesis that specifies the functional form.

The regression is straightforward. Two other tasks were not. Standardising the supernovae by the Phillips width–luminosity relation cut the scatter from 0.4 to 0.15 mag, a factor of 2.7 in the noise and a factor of seven in the number of events needed. Assembling supernovae at $z = 0.3$ – 0.8 put observations where the accelerating and

decelerating models differ by 0.2–0.3 mag. Neither task is statistical.

The estimates of both groups, under flatness, give

$$\begin{aligned}\hat{q}_0 &= \frac{\hat{\Omega}_M}{2} - \hat{\Omega}_\Lambda = 0.14 - 0.72 = -0.58, \\ z_t &\approx 0.73, \\ \hat{\Omega}_\Lambda &= 0.72 \pm 0.09 \text{ (stat)}.\end{aligned}$$

Roughly fifty supernovae sufficed because the Friedmann equations fixed the functional form and left three parameters. A learner obliged to discover the form from data would need thousands of events and would still not know what it had found. Theory compresses the sample size that data must supply.

The 1998 fits estimated parameters. The model’s prediction that supernovae beyond $z \approx 1.4$ would appear brighter than an empty universe estimated nothing, and [Riess et al. \[2004\]](#) confirmed it (Section 11). No systematic-error model produced that change of sign.

The supernovae measured one function, $d_L(z)$, and one combination of two parameters within it. Separating Ω_M from Ω_Λ required the CMB, whose constraint runs nearly orthogonal to theirs. Distinguishing a cosmological constant from a rolling field will require the growth rate of structure, which no one would measure without a theory predicting that the two differ. Theory sets the hypothesis class. Theory then names the measurement that discriminates within it.

Riess, Schmidt, and Perlmutter measured the curvature of the Hubble diagram and found $\Omega_\Lambda \approx 0.72$. They did not explain what dark energy is. Quantum field theory still predicts a vacuum energy some 10^{120} times larger [[Weinberg, 1989](#)].

A The Deceleration Parameter and the Second Friedmann Equation

The deceleration parameter q_0 follows from the second Friedmann equation. With matter (pressure $p_M = 0$) and cosmological constant ($p_\Lambda = -\rho_\Lambda c^2$):

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda}{3} = -\frac{4\pi G \rho_M}{3} + \frac{\Lambda}{3}. \quad (34)$$

Evaluating at $t = t_0$ (today, $\rho_{M,0} = 3H_0^2 \Omega_M / (8\pi G)$, $\Lambda = 3H_0^2 \Omega_\Lambda$):

$$\frac{\ddot{a}_0}{a_0} = -\frac{H_0^2 \Omega_M}{2} + H_0^2 \Omega_\Lambda. \quad (35)$$

Therefore

$$q_0 = -\frac{\ddot{a}_0 a_0}{\dot{a}_0^2} = -\frac{\ddot{a}_0 / a_0}{H_0^2} = \frac{\Omega_M}{2} - \Omega_\Lambda. \quad (36)$$

For $\Omega_M < 2\Omega_\Lambda$ the universe is accelerating today. The best-fit flat model ($\Omega_M = 0.28$, $\Omega_\Lambda = 0.72$) gives $q_0 = 0.14 - 0.72 = -0.58$: the universe currently accelerates at a rate $|q_0| = 0.58$ times the Hubble deceleration.

B Low-Redshift Expansion and the Hubble Diagram

At low redshifts, the luminosity distance has a Taylor expansion in z :

$$d_L(z) = \frac{cz}{H_0} \left[1 + \frac{1}{2}(1 - q_0)z + O(z^2) \right]. \quad (37)$$

The distance modulus then expands as

$$\mu(z) = 5 \log_{10} \left(\frac{cz}{H_0 \text{ Mpc}} \right) + 25 + \frac{5(1 - q_0)}{2 \ln 10} z + O(z^2). \quad (38)$$

The coefficient $5/(2 \ln 10) = 1.086$, so the cosmological term is $1.086(1 - q_0)z$. At $z = 0.1$ and $q_0 = -0.58$ it equals 0.17 mag, which exceeds $\sigma_{\text{int}} \approx 0.15$ mag. The term is not negligible, and the low- z sample is not free of cosmological information.

Leverage defeats the low- z sample. The coefficient multiplies z , and the Calán/Tololo events span only $z \in [0.01, 0.10]$. For 29 events spread over that range, $\sqrt{S_{zz}} = 0.145$,

so the standard error on the coefficient is $\sigma/\sqrt{S_{zz}} = 0.17/0.145 = 1.17$, and

$$\text{SE}(q_0) = \frac{1.17}{1.086} \approx 1.1. \quad (39)$$

A standard error of 1.1 on a parameter whose competing values are -0.58 and $+0.5$ settles nothing. The low- z sample constrains $\mathcal{M}$ through Hubble’s law and leaves q_0 open.

The same expansion prices the alternative. Moving to $z = 0.5$ multiplies the signal by five and multiplies $\sqrt{S_{zz}}$ by more, because the leverage grows with the spread of z and not with its level. Millikan’s frequency-range argument has the same form, and the same consideration drove both teams’ observing strategy.

C The Equation of State of Dark Energy

The cosmological constant Λ corresponds to a fluid with equation of state $w_\Lambda = p_\Lambda/(\rho_\Lambda c^2) = -1$: its pressure is equal and opposite to its energy density. A dark-energy fluid with $w \neq -1$ (e.g., a rolling scalar field ϕ with potential $V(\phi)$) also has $\ddot{a} > 0$ provided $w < -1/3$.

The most general measurement equation, allowing w as a free parameter, is

$$d_L(z; \Omega_M, \Omega_{\text{de}}, w) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_M(1+z')^3 + \Omega_{\text{de}}(1+z')^{3(1+w)}}}. \quad (40)$$

The 1998–1999 analyses assumed $w = -1$. Later surveys—SNLS, Union, JLA, DES—relaxed it [Conley et al., 2011, Sullivan et al., 2011]. Combined supernova, CMB and baryon-acoustic-oscillation data now give $w = -1.03 \pm 0.03$, consistent with a cosmological constant. That figure comes from a secondary source and should be checked against the primary analyses.

The precision on w understates how little it settles. A rolling scalar field with w within a few percent of -1 over the observed redshift range is indistinguishable from Λ by these data, and so are modified-gravity models tuned to reproduce the same expansion history. Supernovae measure $d_L(z)$. Any theory delivering the same $d_L(z)$ is observationally identical to any other, and the remaining uncertainty in w now comes from systematics rather than photon statistics.

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